Michiel De Wilde



Contact Information
Email: michiel.dewilde@ist.ac.at
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Citizenship: Belgium

Research interests

Mathematical physics: Quantum many-body-systems, Topological phases
Side project: Astrophysics: stellar and solar flares.

Education

2025 – present **ISTA** – Kosterneuburg, Austria
PhD with Robert Seiringer

2023 – 2025 **KU Leuven** – Leuven, Belgium
Master in Mathematics *Summa cum Laude*
Project: Classification of Symmetry Protected Topological Phases in 2D
Mentor: Professor Wojciech De Roeck.

2020 – 2023 **KU Leuven** – Leuven, Belgium
Bachelor in Physics, *Magna cum laude.*
Project: Entanglement and topology in quantum spin systems: Toric code.
Mentor: Dr. Bruno de Oliveira Carvalho.

2020 – 2023 **KU Leuven** – Leuven, Belgium
Bachelor in Mathematics, *Magna cum laude*
Project: Optically thick Radiative Transfer in stars
Mentor: Professor Malcolm Druett.

Publications

2026 **Synthesizing Sun-as-a-star flare spectra from high-resolution solar observations (Corrigendum)**
M. De Wilde, A. G. M. Pietrow, M. K. Druett, A. Pastor Yabar, J. Koza, I. Kontogiannis, O. Andriienko, A. Berlicki, A. R. Brunvoll, J. de la Cruz Rodríguez, J. Thoen Faber, R. Joshi, D. Kuridze, D. Nóbrega-Siverio, L. H. M. Rouppe van der Voort, J. Rybák, E. Scullion, A. M. Silva, Z. Vashalomidze, A. Vicente Arévalo, A. Wiśniewska, R. Yadav, T. V. Zaqarashvili, J. Zbinden, E. S. Øyre.
Astronomy & Astrophysics.

2026 [under review] **On the effects of spherical geometry on coronal mass ejection detection**
A. G. M. Pietrow, M. De Wilde, M. Druett, H. Eklund, J. D. Alvarado-Gomez, V. Fedun.
Nature Astronomy.

2026 **Classification of Locality Preserving Symmetries on Spin Chains**
Alex Bols, Wojciech De Roeck, Michiel De Wilde, Bruno de O. Carvalho.
Communications in Mathematical Physics.

2026 **Arbitrary harmonic functions as Bose–Einstein condensates**
Michiel De Wilde, Robert Seiringer.
Journal of Mathematical Physics.

2026 [preprint] **A complete classification of 2d symmetry protected states with symmetric entanglers**
Alex Bols, Wojciech De Roeck, Michiel De Wilde, Bruno de O. Carvalho.
arXiv.

2025 **Synthesizing Sun-as-a-star flare spectra from high-resolution solar observations**
M. De Wilde, A. G. M. Pietrow, M. K. Druett, A. Pastor Yabar, J. Koza, I. Kontogiannis, O. Andriienko, A. Berlicki, A. R. Brunvoll, J. de la Cruz Rodríguez, J. Thoen Faber, R. Joshi, D. Kuridze, D. Nóbrega-Siverio, L. H. M. Rouppe van der Voort, J. Rybák, E. Scullion, A. M. Silva, Z. Vashalomidze, A. Vicente Arévalo, A. Wiśniewska, R. Yadav, T. V. Zaqarashvili, J. Zbinden, E. S. Øyre.
Astronomy & Astrophysics.

Talks

24/03/2026 **Mathematical Physics & Analysis Seminar (ISTA)**
Title: Arbitrary harmonic functions as Bose-Einstein condensates

13/03/2026 **Solar-stellar and star-planet connections: Mind the gap (Royal Astronomical Society)**
Title: Synthesizing Sun-as-a-star flare spectra from high-resolution solar observations

15/10/2025 **MIXed Colloquium (ISTA)**
Title: Classification of locality-preserving symmetries on spin chains

Internship

2023 – 2025 **Using spatially resolved solar flare data to assess spectral signatures of physical phenomena in stellar data**
Mentors: Malcolm Druett and Alexander Pietrow.
Work is to be published in paper above.

Volunteer Work and Commitment

2020 – Present **vzw Pagadderke (board member)**
Served as a board member for Pagadderke vzw, a non-profit dedicated to providing vacation experiences and building networks for children living in poverty. Managed large-scale projects and was responsible for securing grant funding.

2018 – Present **vzw Pagadderke (volunteer)**
volunteer at weekends/vacations/daytrips of vzw Pagadderke.

2020 – present **Bizon vzw (Camp Counselor)**
Led summer camp programs for children from marginalized communities and socially vulnerable contexts, providing a safe and supportive environment for outdoor activities and personal development. Led a team of different background and handled crisis situations associated to the target group.

2019 – pressent **Bizon vzw (volunteer)**
Animator on summer camp programs of Bizon vzw. Advising on subsidy files.

2020 – pressent **Kamino vzw (Camp Counselor)**
Led summer camp programs and organized weekends for children at Kamino.

2024 – present **Omkaderd wonen (atmosphere management)**
Omkaderd wonen KU Leuven is a program where students live together with and take care of students with disabilities. 8 hours per week Permance for every member and atmosphere mangement is responsible for teambuilding and socializing.

2021 **Cursus hoofd animator** (Formation Camp Counselor) at IJD

2017 **Cursus animator** (Formation animator in youth work) at IJD

Technical skills

Programming languages
Proficient in: Python
Familiar with: R, sage, matlab

Languages
Dutch (Native), English (Fluent), French (Conversational), German (Beginner)

Olympiads

Flemish Math Olympiad (4x), Flemish Physics Olympiad (2x), Flemish Biology Olympiad (2x), Flemish Chemistry Olympiad (2x), Flemish Geography Olympiad (2x), Flemish astronomy Olympiad (1x)

Other interests

Reading, sailing, laughing, basketball